

Outside Good Uncertainty

Ordinal Dual Response in Choice-Based Conjoint Analysis

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We provide a behavioral model in which consumers can identify their most-preferred option among a set of alternatives but face uncertainty in their evaluation of the outside good. The model rationalizes the dual-response format used in choice-based conjoint analysis and extends the follow-up purchase question from binary to ordinal, accommodating scales with any number of points. The resulting likelihood couples a multinomial logit for the forced choice with a cumulative-link ordinal model driven by the task’s inclusive value, and it nests the leading binary dual-response models as its two-point special case. We derive the information gains from finer response scales, provide estimation routines for aggregate and hierarchical Bayesian versions of the model — including an extension with respondent-specific cut points — and demonstrate the model’s superior performance relative to the dichotomization and constant-probability heuristics currently used by practitioners. An empirical application uses a choice-based conjoint study conducted in partnership with a consumer insights consultancy.

1 Introduction

Consumers are good at telling us which product they like best. They are considerably less certain about whether they would actually buy it. A shopper can pick a preferred bluetooth speaker from a shelf of a dozen — or a preferred color from a dozen options of the same model — and still hesitate at the moment of purchase; industry trackers consistently estimate that roughly seven in ten online shopping carts are abandoned before checkout.¹ The comparison among displayed alternatives and the commitment to purchase are different cognitive acts: the former is a choice among fully described options in front of the consumer, while the latter requires her to weigh the winner against an outside option — keep the money, wait for a sale, buy elsewhere, do without — whose value she has not yet fully resolved. Survey respondents

¹TODO: add citable source. The Baymard Institute maintains a running meta-analysis of cart-abandonment studies with an average of roughly 70%.

reflect this distinction naturally. Asked “which of these would you choose?”, they answer readily; asked “would you buy it?”, many prefer to answer “probably” than to commit to yes or no. Uncertainty about a purchase *feels natural*, and stated purchase-likelihood scales have been a fixture of marketing research for over half a century (Juster 1966; Morrison 1979).

Choice-based conjoint analysis, the workhorse of stated-preference measurement in marketing (Allenby et al. 2019), has developed its own device for separating these two acts: the *dual-response* design. In each task the respondent first makes a forced choice among the displayed product profiles, and then answers a follow-up question about whether she would actually purchase her selection. The design was introduced to solve a practical problem with the traditional no-choice option. When “none of these” appears alongside the profiles in a single-stage task, every task in which it is chosen contributes nothing to the estimation of attribute preferences, and the efficiency loss grows with no-choice incidence (Brazell et al. 2006). The dual response recovers that lost information — the forced choice always reveals preference among profiles — while still measuring demand for the category through the follow-up question, and it does so without systematically biasing preference estimates (Brazell et al. 2006; Dhar and Simonson 2003). Following its implementation in Sawtooth Software’s widely used CBC system (Diener et al. 2006; Sawtooth Software 2017), the dual-response format became standard practice in commercial conjoint research, and it has continued to attract methodological attention, from willingness-to-pay measurement (Schlereth and Skiera 2017) to the recent behavioral and econometric re-examinations of Hung, Liu, et al. (2025) and Hung, Kurz, et al. (2025).

Practice, however, has quietly moved beyond the models available to analyze it. The canonical dual-response question is binary — “would you buy the option you selected above? (yes/no)” — and the canonical models are correspondingly binary: a multinomial logit for the forced choice paired with a binary logit for the purchase question (Diener et al. 2006; Hung, Liu, et al. 2025). But many applied researchers instead field an *ordinal* follow-up — “how likely are you to buy the option you selected above?”, answered on a five- or seven- or eleven-point scale from “very unlikely” to “very likely.” The ordinal format is attractive precisely because purchase uncertainty feels natural to respondents: it lets them express a degree of conviction rather than forcing a resolution they have not reached. It is also, as we show, more informative. Yet there is no statistical model for it. Analysts confronted with ordinal dual-response data today either *dichotomize* the scale at an arbitrary point (top box or top two boxes count as “buy”) and estimate the binary model, or apply *constant-probability weights* that map each scale point to a fixed purchase probability regardless of what was on the screen. The first discards information the researcher paid to collect and hard-codes an arbitrary cut; the second, as our model makes precise, ignores the fact that the meaning of “somewhat likely” depends on how attractive the displayed alternatives were.

This paper provides the missing model. We start from a simple behavioral premise: the consumer can rank the goods in front of her because she observes everything relevant about them, but she faces genuine uncertainty about the outside good — the not-yet-arrived purchase occasion that the follow-up question asks her to imagine. This premise does three things at

once. It *rationalizes the dual-response design*: the forced choice is the question the consumer can answer with certainty, and the follow-up is the question about which she is uncertain, so the design cleanly partitions what she knows from what she does not. It *rationalizes the ordinal format*: a consumer who has not resolved her outside-good uncertainty holds a purchase probability, and an ordinal scale is exactly an instrument for reporting a probability in bins. And it *delivers a likelihood*: because the maximum of Gumbel utilities is itself Gumbel — with location equal to the log-sum inclusive value — and is independent of which alternative attains it, the joint probability of the pair (chosen profile, ordinal report) factors into a multinomial logit times a cumulative-link ordinal model driven by the inclusive value of the task.

Within this framework we develop two variants that differ only in how literally the respondent carries her uncertainty into the report. In Model A she reports the purchase probability itself, binned by cut points on the unit interval; the implied ordinal likelihood has a Gumbel (complementary log-log type) link. In Model B she behaves as the binary literature implicitly assumes — resolving the outside-good draw and grading the resulting utility difference — which yields a standard ordered logit link. The two variants have identical parameter counts and are numerically close, but they carry different behavioral interpretations, and their binary special cases differ in an instructive way. With a two-point scale, Model B collapses *exactly* to the “Unified Model” of Hung, Liu, et al. (2025) — equivalently the DR-AnyMax model of Diener et al. (2006), and, after telescoping, the standard single-stage multinomial logit with a no-choice constant (Haaijer et al. 2001). Our model is thus a strict generalization of the leading binary dual-response model: practitioners who adopt it lose nothing at $W = 2$ and gain the ability to use every scale point of a richer follow-up. Model A’s binary case, by contrast, implies that even the familiar yes/no follow-up should be analyzed with a complementary log-log rather than logistic link when respondents remain genuinely uncertain — a one-line modification of existing practice that is testable on existing data.

We make four contributions. First, the behavioral model itself: a utility-theoretic account of dual-response conjoint in which outside-good uncertainty is the primitive, which rationalizes the design, selects the link function, and nests the existing binary models as the two-point special case. Second, an analysis of the information content of the ordinal follow-up: we derive the Fisher-information increment contributed by a W -point second response and show it is positive semi-definite and non-decreasing in scale refinement, extending the binary efficiency results of Brazell et al. (2006) and Hung, Liu, et al. (2025) and giving design guidance on the choice of W . Third, estimation machinery: maximum likelihood for the aggregate model and a custom Markov chain Monte Carlo sampler for the hierarchical Bayesian model with heterogeneous preferences — including an extension with respondent-specific cut points that accommodates individual differences in scale usage (Greene and Hensher 2010; Falco et al. 2015) — with complete replication code in R. Fourth, evidence: a simulation study demonstrating parameter recovery and quantifying the losses from dichotomization and constant-probability weighting, and an empirical application to a commercial choice-based conjoint study with an ordinal dual response conducted in partnership with a consumer insights consultancy.

Our work sits at the intersection of three literatures. The dual-response and no-choice literature

in marketing established the design’s efficiency rationale and its estimation conventions (Brazell et al. 2006; Diener et al. 2006; Haaijer et al. 2001; Schlereth and Skiera 2017; Hung, Liu, et al. 2025), and recent work shows the outside good remains the format’s active research frontier — including evidence that its utility can drift over the course of a survey (Hung, Kurz, et al. 2025). The opt-out literature in choice modeling catalogs how misspecifying the outside option distorts welfare and share predictions (Campbell and Erdem 2019). And a literature on ordered and mixed response formats in preference measurement — from unified treatments of ratings, rankings, and choices (Marshall and Bradlow 2002) to ordered-choice models with reporting heterogeneity (Greene and Hensher 2010; Falco et al. 2015) — supplies tools we adapt to the dual-response setting. To our knowledge, no prior work provides a utility-consistent joint model of a forced choice and an ordinal purchase evaluation.

The remainder of the paper proceeds as follows. Section 2 develops the behavioral model, derives the joint likelihood, establishes the nesting relationships, and presents the information-content and estimation results. Section 3 reports the simulation study. Section 4 presents the empirical application. Section 5 discusses implications for practice, including guidance on scale design, and Section 6 concludes.

2 Model

2.1 Setup

Consumer $i = 1, \dots, N$ faces choice tasks $t = 1, \dots, T$. Each task presents a set of “inside” goods $\mathcal{J}^+ = \{1, 2, \dots, J\}$ alongside the ever-present “outside” good $j = 0$, so that the full choice set is $\mathcal{J} = \{0, 1, 2, \dots, J\}$.² Consumer i derives utility u_{itj} from good j in task t :

$$u_{itj} = h(\mathbf{x}_{tj}, \boldsymbol{\beta}_i) + \eta_{itj}, \quad (1)$$

where $\mathbf{x}_{tj}$ is a length- P column vector of good characteristics, $\boldsymbol{\beta}_i$ is a length- P column vector of consumer-specific taste parameters, and η_{itj} encapsulates factors known to the consumer but unobserved by the researcher. We take $h(\mathbf{x}_{tj}, \boldsymbol{\beta}_i) = \mathbf{x}_{tj}'\boldsymbol{\beta}_i$ and write $V_{itj} := \mathbf{x}_{tj}'\boldsymbol{\beta}_i$ for this deterministic component, but the linear specification is not required.

The outside good is special in two respects. First, it is associated with a zero vector of characteristics ($\mathbf{x}_{t0} = \mathbf{0}$), so that $V_{it0} = 0$; this is the standard normalization that anchors the location of utility (Haaijer et al. 2001). Second — and this is the central behavioral assumption of the paper — the consumer observes η_{itj} for all inside goods but *does not* observe η_{it0} . The

²We hold J fixed across tasks and consumers to lighten notation; nothing in what follows requires it.

inside goods are described completely on the screen in front of her; the outside good stands in for a purchase occasion that has not yet arrived, and her evaluation of it is uncertain.^{3 4}

In each task the consumer provides two responses:

1. **First response (choice).** She reports her preferred inside good,

$$j_{it}^* = \operatorname{argmax}_{j \in \mathcal{J}^+} u_{itj}. \quad (2)$$

This is a *forced* choice: the outside good is not among the options.

2. **Second response (ordinal purchase evaluation).** She reports a value y_{it} on a discrete ordinal scale $w \in \mathcal{W} = \{1, \dots, W\}$ (e.g., “very unlikely” to “very likely” on a five-point scale) that characterizes her evaluation of purchasing j_{it}^* — that is, of choosing j_{it}^* over the outside good. The binary dual-response format of Brazell et al. (2006) and Diener et al. (2006) is the special case $W = 2$.

We model $\eta_{itj} \stackrel{iid}{\sim} \text{Gumbel}(0, 1)$, independent of $\mathbf{x}_{tj}$ and independent across goods, tasks, and consumers. Two classical properties of the Gumbel family organize everything that follows (McFadden 1981; Cardell 1997).

Lemma 2.1. *Let $u_{itj} = V_{itj} + \eta_{itj}$ with $\eta_{itj} \stackrel{iid}{\sim} \text{Gumbel}(0, 1)$ for $j \in \mathcal{J}^+$, and define $u_{it}^* = \max_{j \in \mathcal{J}^+} u_{itj}$. Then*

- (i) $u_{it}^* \sim \text{Gumbel}(\bar{\mu}_{it}, 1)$ with location

$$\bar{\mu}_{it} = \ln S_{it}, \quad S_{it} := \sum_{j \in \mathcal{J}^+} \exp(V_{itj}); \quad (3)$$

- (ii) *the identity of the maximizer j_{it}^* and the value of the maximum u_{it}^* are statistically independent.*

Property (ii) is what makes the dual-response likelihood tractable: whatever the consumer tells us about the level of her best utility (the second response) is independent of which good delivered it (the first response), conditional on the deterministic utilities.

³This can be motivated by a framework in which $u_{it0} = 0$ and $u_{itj} = \mathbf{x}_{tj}'\beta_i + \eta_{itj} - \eta_{it0}$ for $j \in \mathcal{J}^+$, where η_{it0} captures the consumer’s uncertainty about her future tastes or circumstances at the moment of purchase. Because utilities are ordinal and η_{it0} is a common shock, it plays no role in the comparison among inside goods.

⁴While the parametrization $\eta_{it0} \sim \text{Gumbel}(0, 1)$ preserves symmetry among the $J + 1$ goods and is thus a natural choice, the framework easily accommodates alternative distributions for η_{it0} . For example, an affine function of individual characteristics could accommodate individual-level variation in the propensity to prefer the outside good.

2.2 First Response: Multinomial Logit

The first response is a standard forced choice among the inside goods. Under the Gumbel assumption,

$$\Pr(j_{it}^* = j) = \frac{\exp(V_{itj})}{\sum_{k \in \mathcal{J}^+} \exp(V_{itk})} = \frac{\exp(V_{itj})}{S_{it}}, \quad (4)$$

the multinomial logit over inside goods only. Note that the outside good does not appear: the forced choice is uncontaminated by the consumer's uncertain evaluation of the outside good, which is precisely the design's appeal (Brazell et al. 2006).

2.3 Second Response: An Ordinal Model of the Purchase Evaluation

What does the consumer consult when answering “how likely are you to purchase the option you selected?” We develop two behavioral models that differ only in how the consumer treats the unobserved outside-good shock η_{it0} . Both lead to likelihoods of the same *cumulative-link* form, differing only in the link function, and both depend on the choice set only through the inclusive value $\bar{\mu}_{it}$.

2.3.1 Model A: Reporting a Purchase Probability

Under Model A the consumer takes her uncertainty about the outside good seriously and reports a *probability*. She knows u_{it}^* (she just identified j_{it}^*) but does not know η_{it0} , so from her perspective the probability that she would follow through and purchase j_{it}^* is

$$p_{it} = \Pr(\eta_{it0} < u_{it}^* \mid u_{it}^*) = F(u_{it}^*), \quad (5)$$

where $F(z) = \exp(-e^{-z})$ is the standard Gumbel CDF. The ordinal scale discretizes the unit interval: fix cut points $0 = \alpha_0 < \alpha_1 < \dots < \alpha_{W-1} < \alpha_W = 1$ and suppose the consumer reports

$$y_{it} = w \iff p_{it} \in [\alpha_{w-1}, \alpha_w]. \quad (6)$$

We assume consumers share the definitions of the qualitative scale labels, so the α_w are common across consumers (we relax this in Section 2.7.1); the interior cut points are unobserved by the researcher and are estimated.

Reporting $y_{it} = w$ is equivalent to reporting that $u_{it}^* \in [F^{-1}(\alpha_{w-1}), F^{-1}(\alpha_w)]$, an interval statement about the consumer's best utility. From the researcher's perspective, $u_{it}^* \sim \text{Gumbel}(\bar{\mu}_{it}, 1)$ by Lemma 2.1(i), so

$$\begin{aligned} \Pr(y_{it} = w) &= F_{\text{Gumbel}(\bar{\mu}_{it}, 1)}(F^{-1}(\alpha_w)) - F_{\text{Gumbel}(\bar{\mu}_{it}, 1)}(F^{-1}(\alpha_{w-1})) \\ &= \alpha_w^{S_{it}} - \alpha_{w-1}^{S_{it}}, \end{aligned} \quad (7)$$

using $F_{\text{Gumbel}(\bar{\mu}, 1)}(c) = \exp(-e^{-(c-\bar{\mu})}) = \exp(e^{\bar{\mu}} \ln \alpha) = \alpha^S$ when $c = F^{-1}(\alpha) = -\ln(-\ln \alpha)$. The second-response probabilities are differences of powers of the cut points, with the exponent $S_{it} = e^{\bar{\mu}_{it}}$ summarizing the attractiveness of the task's inside goods: the more attractive the set, the more probability mass shifts toward high scale points.

2.3.2 Model B: Grading a Resolved Comparison

Under Model B the consumer behaves as the binary dual-response literature implicitly assumes: she resolves (or acts as if she resolves) the outside-good shock and evaluates the realized utility difference

$$d_{it} = u_{it}^* - \eta_{it0}.$$

Rather than reporting only its sign — the binary “would you buy it?” — she grades its magnitude on the W -point scale: fix utility-scale cut points $-\infty = c_0 < c_1 < \dots < c_{W-1} < c_W = \infty$ and suppose

$$y_{it} = w \iff d_{it} \in [c_{w-1}, c_w]. \quad (8)$$

From the researcher's perspective, $u_{it}^* \sim \text{Gumbel}(\bar{\mu}_{it}, 1)$ and $\eta_{it0} \sim \text{Gumbel}(0, 1)$ are independent, so their difference is logistic: $d_{it} \sim \text{Logistic}(\bar{\mu}_{it}, 1)$. Hence

$$\Pr(y_{it} = w) = \Lambda(c_w - \bar{\mu}_{it}) - \Lambda(c_{w-1} - \bar{\mu}_{it}), \quad \Lambda(z) = \frac{1}{1 + e^{-z}}. \quad (9)$$

This is an ordered logit (Greene and Hensher 2010) with the inclusive value $\bar{\mu}_{it}$ as the (negatively signed) linear predictor.

2.3.3 A Unified Cumulative-Link Representation

Define $c_w = F^{-1}(\alpha_w)$ in Model A so that both models carry utility-scale cut points. Then both second-response models take the cumulative-link form

$$\Pr(y_{it} \leq w) = G(c_w - \bar{\mu}_{it}), \quad G = \begin{cases} F \text{ (standard Gumbel)} & \text{Model A,} \\ \Lambda \text{ (standard logistic)} & \text{Model B.} \end{cases} \quad (10)$$

The behavioral story pins down the link: a consumer who *carries* her outside-good uncertainty into the report (Model A) generates a Gumbel — i.e., complementary log-log type — link, while a consumer who *resolves* it and grades the outcome (Model B) generates the familiar logistic link. The two links are numerically close over most of their range and diverge in the tails; which better describes respondents is an empirical question we take up in Section 3 and Section 4. Both are estimable at identical parameter counts ($W - 1$ interior cut points), so the comparison is a clean, non-nested horse race between links.

2.4 Joint Likelihood

By Lemma 2.1(ii), the value of u_{it}^* is independent of the identity of j_{it}^* ; and η_{it0} is independent of the inside-good shocks. The two responses in a task are therefore independent conditional on $(\mathbf{x}_t, \beta_i)$, and the per-task likelihood contribution is simply the product of Equation 4 and Equation 10:

$$\ell_{it}(\beta_i, \mathbf{c}) = \underbrace{\frac{\exp(V_{itj^*})}{S_{it}}}_{\text{choice (MNL)}} \times \underbrace{\left[G(c_{y_{it}} - \bar{\mu}_{it}) - G(c_{y_{it}-1} - \bar{\mu}_{it}) \right]}_{\text{ordinal purchase evaluation}}, \quad (11)$$

with $\mathbf{c} = (c_1, \dots, c_{W-1})$ and the full likelihood $L_i = \prod_t \ell_{it}$. We emphasize that this factorization is not an *assumption* of independent errors across the two responses, as in the “DR-2Max” tradition (Diener et al. 2006; Sawtooth Software 2017); it is a *consequence* of the max-stability properties of the Gumbel family within a single, internally consistent utility draw. The same realized utilities generate both responses.

2.5 The Model in Miniature

Before turning to special cases, we make the machinery concrete with the smallest interesting instance of the model: two consumers with *identical* preferences, one attribute, one task each, and a three-point scale. The example is deliberately tiny, but it exhibits the model’s central mechanism — the two responses read two different features of the same utilities — in a form that can be followed with a hand calculator.

Let both consumers share the scalar taste parameter $\beta = 1$, which we treat as known for the arithmetic. Each faces one task with two inside goods described by a single attribute x , answers the forced choice, and then reports on the three-point scale $\mathcal{W} = \{\text{"unlikely"}, \text{"unsure"}, \text{"likely"}\}$ with utility-scale cut points $\mathbf{c} = (c_1, c_2) = (0, 2)$. (For Model A these correspond to probability-scale cut points $(\alpha_1, \alpha_2) = (F(0), F(2)) = (0.37, 0.87)$: the consumer answers “unsure” when her purchase probability lies between 37% and 87%.) The two tasks differ only in the *level* of the attribute, not its spread: consumer 1 faces an attractive pair, $x = (2, 1)$, and consumer 2 an unattractive pair, $x = (-1, -2)$.

Walk through the model’s account of their behavior, giving both consumers the *same* realized shocks $\eta = (0.30, -0.86)$ on their two inside goods. Consumer 1’s utilities are $u = (2.30, 0.14)$: she picks her first good and knows her best utility, $u^* = 2.30$. She cannot resolve the outside-good shock η_0 , so under Model A she reports the bin containing her purchase probability $p = F(2.30) = 0.90$ — “likely.” Consumer 2’s utilities are $u = (-0.70, -2.86)$: she also picks her first good, but her best utility is $u^* = -0.70$, her purchase probability is $F(-0.70) = 0.13$, and she reports “unlikely.” (Under Model B the consumers instead resolve η_0 — say $\eta_0 = 0.10$ — and grade $u^* - \eta_0$ against the same cut points: 2.20 and -0.80 , again “likely” and “unlikely.”) The two consumers made *identical* first-response choices from *identical* shock realizations, yet gave opposite second responses.

The researcher-side probabilities show that this is systematic, not an artifact of the particular draws (Table 1). The forced-choice probabilities are *the same function of the data* for both consumers: the MNL depends only on the within-task difference $x_1 - x_2 = 1$, so $\Pr(\text{choose better good}) = \Lambda(\beta) = 0.73$ in both tasks, and no quantity of forced-choice data could ever distinguish them. The tasks separate only through the inclusive value — $S_1 = e^2 + e^1 = 10.11$ versus $S_2 = e^{-1} + e^{-2} = 0.50$, i.e., $\bar{\mu}_1 = 2.31$ versus $\bar{\mu}_2 = -0.69$ — which the cumulative-link stage reads against the cut points: the ordinal response distributions are near mirror images. (With $c_1 = 0$, consumer 2’s probability of “unlikely” is $\Lambda(-\bar{\mu}_2) = 1/(1 + S_2) = 0.67$ — precisely the familiar no-choice probability of the single-stage MNL, previewing the nesting results below.)

Table 1: Two tasks that are indistinguishable in the forced choice and nearly opposite in the ordinal report ($\beta = 1$, $\mathbf{c} = (0, 2)$, $W = 3$).

	Consumer 1 (attractive task)	Consumer 2 (unattractive task)
Attribute values x	(2, 1)	(−1, −2)
$\Pr(\text{choose better good})$	0.73	0.73
$S = \sum_j e^{V_j}$	10.11, 2.31	0.50, −0.69
$\bar{\mu} = \ln S$		
$\Pr(y = w)$, Model B	(0.09, 0.33, 0.58)	(0.67, 0.27, 0.06)
$\Pr(y = w)$, Model A	(0.00, 0.25, 0.75)	(0.60, 0.33, 0.07)
$E[p \mid y = \text{"unsure"}]$, Model A	0.80	0.60

The joint likelihood of the observed data — each consumer chose her better good; $y_1 = \text{"likely"}$, $y_2 = \text{"unlikely"}$ — follows from Equation 11 (shown here for Model B):

$$L(\beta, c_1, c_2) = \underbrace{\Lambda(\beta) [1 - \Lambda(c_2 - \bar{\mu}_1(\beta))]}_{\text{consumer 1: } 0.73 \times 0.58} \times \underbrace{\Lambda(\beta) \Lambda(c_1 - \bar{\mu}_2(\beta))}_{\text{consumer 2: } 0.73 \times 0.67}, \quad (12)$$

with $\bar{\mu}_1(\beta) = \beta + \ln(1 + e^\beta)$ and $\bar{\mu}_2(\beta) = -2\beta + \ln(1 + e^\beta)$. The example distills three general lessons.

First, *the two responses divide the labor of identification*. The forced-choice factors read only utility **differences** — they identify β from within-task attribute variation. The ordinal factors read utility **levels**, locating each task’s $\bar{\mu}(\beta)$ against the cut points. All information about demand — whether these consumers would actually buy — flows through the second response; all information about relative preference flows through the first. This is the division that Proposition 2.1 formalizes: with repeated observations of tasks like these, (β, c_1, c_2) is identified because the stacked attribute vector $(2, 1, -1, -2)'$ is not constant across the goods shown; shift every x by a common amount and the choice factors in Equation 12 are untouched while the shift is absorbed one-for-one by the cut points.

Second, *the meaning of a scale label is set-dependent*. The label “unsure” is fixed for the consumer — by construction it means $p \in [0.37, 0.87)$ — but what the researcher should infer from it depends on the task: the expected purchase probability of a consumer who answers “unsure” is 0.80 in the attractive task and 0.60 in the unattractive one (Table 1, last row).⁵ Any procedure that assigns a fixed purchase probability to each scale point — the constant-probability heuristic of Section 2.6 — is therefore internally inconsistent under the model, and the error compounds when calibrations struck on one design are transported to another.

Third, *the links agree where tasks are ordinary and part ways where they are extreme*. Models A and B produce nearly identical distributions for the unattractive task but visibly different ones for the attractive task (Table 1), a miniature of the general point from Section 2.8 that discriminating between the links requires designs that vary task attractiveness.

2.6 Special Cases and Relation to Existing Models

Binary dual response, Model B. Set $W = 2$ with the single cut point $c_1 = \gamma$. From Equation 9, $\Pr(\text{would buy}) = \Lambda(\bar{\mu}_{it} - \gamma) = S_{it}/(e^\gamma + S_{it})$, and the joint probabilities from Equation 11 telescope:

$$\Pr(j, \text{buy}) = \frac{\exp(V_{itj})}{e^\gamma + S_{it}}, \quad \Pr(j, \text{not buy}) = \frac{\exp(V_{itj})}{S_{it}} \cdot \frac{e^\gamma}{e^\gamma + S_{it}}. \quad (13)$$

With $\gamma = 0$ these are *exactly* the joint probabilities of the “Unified Model” of Hung, Liu, et al. (2025) (their Eq. 4), which is in turn algebraically identical to the “DR-AnyMax” model of Diener et al. (2006); and Equation 13 with free γ is the standard single-stage MNL with a no-choice alternative-specific constant (Haaïjer et al. 2001). Our ordinal model is thus a strict generalization of the leading binary dual-response model: at $W = 2$, Model B *is* that model, and for $W > 2$ the additional cut points are the only new parameters.

Binary dual response, Model A. Set $W = 2$ in Equation 7: $\Pr(\text{not buy}) = \alpha_1^{S_{it}} = F(c_1 - \bar{\mu}_{it})$, a *complementary log-log* rather than logistic binary response. Model A therefore implies that even the familiar binary dual response is misspecified — mildly, given the proximity of the links — when consumers genuinely face outside-good uncertainty at the moment they answer. This observation gives the binary special case of Model A independent interest: it provides a one-parameter alternative to the binary logit second stage that can be tested on existing binary dual-response data.

DR-2Max. The other binary model in the literature, “DR-2Max” (Diener et al. 2006), as implemented in Sawtooth Software’s CBC/HB (Sawtooth Software 2017), models the second response as a binary logit on the *chosen good’s* deterministic utility: $\Pr(\text{buy} | j^*) =$

⁵The conditional expectation is $E[F(u^*) | u^* \in [c_1, c_2)]$ with $u^* \sim \text{Gumbel}(\bar{\mu}, 1)$, which has the closed form $\frac{S}{S+1} \left[e^{-(S+1)e^{-c_2}} - e^{-(S+1)e^{-c_1}} \right] / \left[e^{-Se^{-c_2}} - e^{-Se^{-c_1}} \right]$.

$e^{V_{itj^*}} / (e^{V_{itj^*}} + e^\gamma)$. This requires drawing a fresh error for the chosen good in the second stage — the consumer who just declared j^* best re-evaluates it from scratch. Under our framework such conditioning is unnecessary: by Lemma 2.1(ii) the distribution of the best utility does not depend on which good attained it, so the second response should depend on the set only through $\bar{\mu}_{it}$. This distinction is testable, and the existing evidence favors the inclusive-value form: Diener et al. (2006) found DR-AnyMax fit dual-response data better than DR-2Max, and Hung, Liu, et al. (2025) reach the same conclusion for their Unified Model.

Practitioner heuristics for ordinal dual responses. When practitioners field an ordinal purchase-likelihood question today, common practice is either (i) *dichotomization* — collapse the scale at some point (e.g., top box or top two boxes) and estimate the binary model, discarding the ordinal information and hard-coding an arbitrary cut; or (ii) *constant-probability weighting* — map each scale point to a fixed purchase probability (e.g., “definitely would buy” = 0.9) independent of the choice set. Both are nested within, or dominated by, Equation 11: dichotomization is Equation 13 applied to a coarsened y_{it} , and constant-probability weighting violates Equation 7, under which the mapping from scale point to purchase probability necessarily depends on the attractiveness of the inside goods through S_{it} . These two heuristics are the benchmarks in our simulation study (Section 3).

2.7 Heterogeneity

We accommodate preference heterogeneity with a standard hierarchical Bayesian specification (Rossi et al. 2005; Train 2001):

$$\beta_i \stackrel{iid}{\sim} \mathcal{N}(\bar{\beta}, \Sigma_\beta), \quad (14)$$

with conjugate priors on $(\bar{\beta}, \Sigma_\beta)$. In the main model the cut points $\mathbf{c}$ are common across consumers: the scale labels (“very likely,” “somewhat likely,” ...) are assumed to mean the same thing to everyone, while *preferences* vary.

2.7.1 Heterogeneous Cut Points

The assumption of a shared reporting scale is convenient but contestable: a rich literature documents individual differences in the use of ordinal response scales (Greene and Hensher 2010; Falco et al. 2015). We therefore develop an extension with consumer-specific cut points. Parameterize

$$c_{i1} \in \mathbb{R}, \quad c_{iw} = c_{i,w-1} + \exp(\delta_{iw}) \text{ for } w = 2, \dots, W-1, \quad (15)$$

and let $\theta_i = (c_{i1}, \delta_{i2}, \dots, \delta_{i,W-1})'$ follow a multivariate normal population distribution, estimated jointly with (and, if desired, correlated with) β_i . The exponentiated increments enforce monotonicity without constrained sampling. Identification at the individual level is necessarily partial — with T tasks per consumer, individual cut points and individual preferences are

jointly informed by only T ordinal reports — so the hierarchical prior does real work here, shrinking individual scale usage toward the population norm. Section 3 examines both the payoff to modeling cut-point heterogeneity when it is present in the data-generating process and the robustness of the common-cut-point model when it is not.

2.8 Identification

Because the second response depends on the choice set only through $\bar{\mu}_{it}$, and because the cut points sit on the same utility scale as $\bar{\mu}_{it}$, the model requires — and, given a suitable design, requires only — a location normalization. This subsection makes the requirement precise. Fix the aggregate model ($\beta_i = \beta$) and let the data consist of tasks $t = 1, \dots, n$ with inside-good attribute vectors $\{\mathbf{x}_{tj}\}_{j \in \mathcal{J}^+}$ and responses (j_t^*, y_t) generated by Equation 11 with parameter $\theta = (\beta, \mathbf{c}) \in \mathbb{R}^P \times \mathcal{C}$, where $\mathcal{C} = \{\mathbf{c} : c_1 < \dots < c_{W-1}\}$. Call θ and θ' *observationally equivalent* if they imply the same joint probabilities $\Pr_\theta(j_t^* = j, y_t = w)$ for every task design in the support of the design distribution, and let X denote the design matrix that stacks the rows $\mathbf{x}'_{tj}$ over all inside goods and tasks in that support.

Proposition 2.1. $\theta = (\beta, \mathbf{c})$ is identified if and only if the augmented matrix $[X \ \iota]$ has full column rank $P + 1$, where ι is a conformable vector of ones. Equivalently: the only $(v, \delta) \in \mathbb{R}^P \times \mathbb{R}$ satisfying $\mathbf{x}'_{tj}v = \delta$ for every inside good j and task t is $(v, \delta) = (\mathbf{0}, 0)$.

Proof. (Sufficiency.) Suppose θ and $\theta' = (\beta', \mathbf{c}')$ are observationally equivalent. Marginalizing the joint probabilities over y equates the MNL choice probabilities task by task, which implies $(\mathbf{x}_{tj} - \mathbf{x}_{tk})'(\beta - \beta') = 0$ for all j, k, t ; hence $\mathbf{x}'_{tj}(\beta - \beta') = a_t$, a constant within each task. Marginalizing over j equates the cumulative ordinal probabilities: $G(c_w - \bar{\mu}_t(\beta)) = G(c'_w - \bar{\mu}_t(\beta'))$ for all w, t . Since G is strictly increasing (for both links in Equation 10), $c_w - c'_w = \bar{\mu}_t(\beta) - \bar{\mu}_t(\beta')$ for all w, t ; the left side is free of t and the right side free of w , so both equal a constant δ . Direct computation gives $\bar{\mu}_t(\beta') = \ln \sum_j \exp(V_{tj} - a_t) = \bar{\mu}_t(\beta) - a_t$, so $a_t = \delta$ for every t ; that is, $\mathbf{x}'_{tj}(\beta - \beta') = \delta$ for all j, t . The rank condition then forces $\beta = \beta'$ and $\delta = 0$, hence $\mathbf{c} = \mathbf{c}'$. *(Necessity.)* If some $(v, \delta) \neq (\mathbf{0}, 0)$ satisfies $\mathbf{x}'_{tj}v = \delta$ everywhere, then $\theta' = (\beta + v, \mathbf{c} + \delta\iota)$ shifts every V_{tj} and every cut point by δ , leaving the MNL probabilities (a within-task constant drops out) and the ordinal probabilities ($G(c_w + \delta - (\bar{\mu}_t + \delta))$) unchanged, so $\theta' \neq \theta$ is observationally equivalent to θ . $\square$

Several remarks unpack the content of Proposition 2.1.

What the condition rules out. A violating direction (v, δ) is an attribute combination whose value is the same for every inside good ever shown. The leading case is an intercept: a column of ones in X makes $[X \ \iota]$ rank deficient by construction. The subtler case is a *hidden* intercept: if an attribute is dummy-coded with all L of its levels included, every alternative carries exactly one of them, so the dummies sum to ι and the same invariance appears (shift all L part-worths and all cut points equally). Attributes must therefore be coded against omitted

reference levels (or effects-coded), and the design must contain no global constant. Notably, this requirement is *stricter* than in the single-stage MNL with a no-choice alternative, where a full set of brand dummies is identified because the no-choice row of zeros breaks the constancy (Haaijer et al. 2001). In the dual-response model the outside good never appears as a design row — its role is absorbed by the cut points — so anything the design holds constant across inside goods is absorbed along with it.

What the normalizations do. The convention $\mathbf{x}_{t0} = \mathbf{0}$ with a Gumbel(0, 1) shock fixes the origin against which the cut points are measured: an outside-good alternative-specific constant would shift all cut points equally and is exactly the invariance in the proof, so it is subsumed rather than restricted. (Its visible trace is the binary case: at $W = 2$, Model B’s single cut point c_1 is the no-choice constant of Equation 13.) The unit Gumbel scale fixes the scale of β , as in any logit. Beyond these, the model imposes a genuine restriction rather than a normalization: the coefficient on $\bar{\mu}_{it}$ in Equation 10 is fixed at 1 by the theory — both responses are driven by the same utility draw — so estimating a free slope λ on $\bar{\mu}_{it}$ and testing $\lambda = 1$ is an overidentification test of the behavioral model, not a required normalization.

Cut points versus the link. Given β (identified from the forced choice alone whenever X ’s within-task differences have full rank), the cut points are exactly identified even *without* variation in task attractiveness: $c_w = \bar{\mu} + G^{-1}(\Pr(y \leq w))$ pointwise. Variation in $\bar{\mu}_{it}$ across tasks therefore overidentifies $\mathbf{c}$ — and it is precisely what identifies the *link*: at a single value of $\bar{\mu}$, any two strictly increasing links fit the W category frequencies perfectly by relabeling cut points, so Models A and B are distinguishable only through how the ordinal distribution shifts as S_{it} varies. Designs intended to compare the links should vary task attractiveness deliberately.

Population versus sample. Under Proposition 2.1 every scale category has strictly positive probability in every task (G is strictly increasing on $\mathbb{R}$ and the cut points are finite), so no support condition is needed for point identification. In finite samples, however, rarely used categories leave their adjacent cut points weakly identified, and near-violations of the rank condition (attribute combinations *nearly* constant across inside goods) manifest as an ill-conditioned information matrix. Both pathologies are visible in the smallest eigenvalues of the observed information, a diagnostic we report alongside the simulation results in Section 3.

The hierarchical model. With heterogeneous β_i , the argument applies respondent by respondent: individual-level identification requires the rank condition on each respondent’s own stacked design (their $T \times J$ rows), which rotating commercial designs typically satisfy. The hierarchy then supplies shrinkage, not identification of the model’s fixed features. In the heterogeneous-cut-point extension (Section 2.7.1) the same location tension reappears at the individual level — a respondent-specific utility shift trades off against c_{i1} within that respondent’s likelihood — and is resolved by the same condition applied to the individual design, with the hierarchical prior doing real work when T is small.

Numerical verification. We verify Proposition 2.1 computationally (replication code accompanies the paper). Data simulated from the raw behavioral process — Gumbel utility

draws, argmax, and the two reporting rules, not the derived likelihood — at three distinct parameter configurations for each model, on designs satisfying the rank condition, are recovered by maximum likelihood: across all configurations the true parameter vector lies inside the 95% likelihood-ratio confidence region, the observed information is well-conditioned, repeated-sampling bias is statistically indistinguishable from zero, and model-based standard errors match the empirical sampling variation. Appending an intercept column — the canonical violation — produces exactly the predicted failure: $[X \ \iota]$ loses rank, the smallest eigenvalue of the observed information collapses to zero, and different starting values converge to different raw $(\beta_{\text{const}}, \mathbf{c})$ on the flat ridge while agreeing, to numerical precision, on the identified combinations $c_w - \beta_{\text{const}}$ and the non-constant coordinates of β . The same exercise confirms the factorization in Equation 11: the empirical joint distribution of (j^*, y) from one million behavioral draws is statistically indistinguishable from the product form.

2.9 Specification Tests

Because the theory restricts more than it parameterizes, the model is overidentified in two directions that convert directly into one-degree-of-freedom specification tests. Both are computed from the aggregate likelihood at trivial cost, and both have well-behaved operating characteristics (Section 3).

The unit-slope test. The behavioral model fixes the coefficient on the inclusive value in the second stage at exactly one — the same utility draw generates both responses — so the natural one-parameter enlargement

$$\Pr(y_{it} \leq w) = G(c_w - \lambda \bar{\mu}_{it}) \quad (16)$$

nests the model at $\lambda = 1$. A free λ is identified by how the ordinal response distribution shifts as task attractiveness varies (Section 2.8), and departures are interpretable: $\lambda < 1$ indicates the second response under-reacts to the quality of the displayed set (consistent with inattention to the set or with scale-use compression), while $\lambda > 1$ indicates over-reaction. Testing $H_0: \lambda = 1$ by likelihood ratio or Wald is an overidentification test of the joint utility account, not a normalization check.

The conditioning test. The framework implies the second response depends on the choice set *only through* $\bar{\mu}_{it}$ (Lemma 2.1), whereas the DR-2Max tradition conditions on the chosen alternative’s utility. The enlargement

$$\Pr(y_{it} \leq w) = G(c_w - [\bar{\mu}_{it} + \delta (V_{itj^*} - \bar{\mu}_{it})]) \quad (17)$$

spans the two: $\delta = 0$ is our model and $\delta = 1$ puts the full weight on the chosen option’s deterministic utility, the DR-2Max predictor. Both points are testable against the estimated δ , so the long-standing 2Max-versus-AnyMax question (Diener et al. 2006; Hung, Liu, et al. 2025) becomes a parameter estimate rather than a model-selection exercise.

We apply both tests to the empirical data in Section 4. Their finite-sample size and power — including full power of the δ test against DR-2Max data at modest sample sizes — are documented in the simulation study.

2.10 The Information Content of the Ordinal Response

How much does a W -point second response add over a binary one? The question has a clean answer in terms of Fisher information, extending the efficiency analysis of Hung, Liu, et al. (2025) from binary to ordinal second responses.

The per-task Fisher information about β decomposes into the familiar MNL term (Huber and Zwerina 1996) plus a second-response increment. Writing $\pi_w = G(c_w - \bar{\mu}) - G(c_{w-1} - \bar{\mu})$ and $g = G'$, the second response contributes

$$\mathcal{I}_{it}^{(2)} = \omega(\bar{\mu}_{it}; \mathbf{c}) \bar{\mathbf{x}}_{it} \bar{\mathbf{x}}'_{it}, \quad \omega = \sum_{w=1}^W \frac{[g(c_w - \bar{\mu}) - g(c_{w-1} - \bar{\mu})]^2}{\pi_w}, \quad (18)$$

where $\bar{\mathbf{x}}_{it} = \sum_{j \in \mathcal{J}^+} \Pr(j_{it}^* = j) \mathbf{x}_{tj}$ is the choice-probability-weighted average attribute vector (the gradient of $\bar{\mu}_{it}$ with respect to β). Two properties follow. First, the increment is positive semi-definite: the second response never destroys information, echoing the binary result of Hung, Liu, et al. (2025). Second, ω is non-decreasing under refinement of the scale partition — coarsening an ordinal response is a garbling in the sense of sufficiency, so the binary response ($W = 2$) is the *least* informative member of the family, and information rises toward the continuous-report limit as $W \rightarrow \infty$. The practical magnitudes — how much of the binary-to-continuous gap a five- or seven-point scale captures, and how the gains vary with task attractiveness — are quantified in Section 3.

2.11 Estimation

The aggregate (homogeneous- β) model is estimated by maximum likelihood: the log of Equation 11 summed over consumers and tasks is smooth in $(\beta, \mathbf{c})$ after the monotonicity-preserving reparameterization of the cut points, and standard quasi-Newton methods converge quickly.

The hierarchical model is estimated with a custom Markov chain Monte Carlo sampler, implemented in R with computational kernels in C++ (Rcpp), following the hybrid Gibbs architecture of Rossi et al. (2005): (1) a random-walk Metropolis step for each β_i (and θ_i in the heterogeneous-cut-point extension) using the individual likelihood $\prod_t \ell_{it}$; (2) a random-walk Metropolis step for the common cut-point block on the unconstrained scale; and (3) conjugate draws for the population moments $(\bar{\beta}, \Sigma_\beta)$. Model comparison uses the log marginal density approximated by the method of Newton and Raftery (1994), matching the practice in Hung, Liu, et al. (2025) and permitting direct comparisons among Model A, Model B, the binary

dual-response models, and the practitioner heuristics. Complete replication code, together with three worked-example notebooks that build from the simplest homogeneous two-attribute case to the full model with heterogeneous cut points, accompanies the paper.

3 Simulation Study

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The simulation program serves six purposes: verify that the estimators recover the model’s parameters and that the identification theory of Section 2.8 bites where it says it does; quantify what dichotomization costs, in the aggregate and — where it matters more — at the individual level; establish the operating characteristics of the specification tests of Section 2.9; probe robustness to the misspecifications a referee should worry about; turn the information results of Section 2.10 into design guidance; and attach economic stakes to the choice of estimator through a pricing counterfactual. All data are simulated from the *behavioral process* — raw Gumbel utility draws, the argmax, and the reporting rules — never from the derived likelihood, so parameter recovery validates the derivation and the code simultaneously. Every experiment is seeded and reproducible from the scripts in the replication archive.

3.1 Design

The aggregate experiments use pooled tasks with $J = 4$ alternatives described by two 3-level attributes (dummy-coded against reference levels) and a continuous price-like attribute ($P = 5$), a $W = 5$ scale, and both links, across four parameter configurations that vary the location and spread of the response distribution — including a *skewed* configuration with realistic bottom-heavy purchase intent (top-2-box share 18.4%). The hierarchical experiments use $N = 300$ respondents, $T = 12$ estimation tasks plus 2 holdouts, $\beta_i \sim \mathcal{N}(\bar{\beta}, \Sigma)$ with substantial heterogeneity, and the same attribute scheme.

3.2 Parameter Recovery and Identification

Aggregate. Three parameter configurations per link, each simulated once at 20{,}000 tasks and estimated by maximum likelihood: in every case the true parameter vector lies inside the 95% likelihood-ratio confidence region, the observed information is strongly positive definite, and the probability-scale cut points of Model A recover to within a percentage point (e.g., a true $\alpha_1 = 0.02$ in the stress configuration is estimated at 0.0196). Repeated-sampling behavior matches the asymptotics: across 40 independent datasets per link at $n = 5,000$, no parameter shows significant bias (max $|t| = 2.05$ over 18 tests) and model-based standard errors match the empirical sampling variation (mean calibration ratio 1.008); quadrupling n shrinks the largest observed bias in proportion, as $O(1/n)$ requires.

The identification condition is sharp. Appending an intercept — the canonical violation of Proposition 2.1 — makes the rank check fail, collapses the smallest eigenvalue of the observed information from roughly 10^3 to 10^{-6} , and produces exactly the predicted flat ridge: different starting values converge to different raw $(\beta_{\text{const}}, \mathbf{c})$ with identical likelihoods, while agreeing to four decimal places on the identified combinations $c_w - \beta_{\text{const}}$ and on every non-constant coordinate of β .

The factorization is exact. The empirical joint distribution of (first response, second response) over one million behavioral draws of a fixed task is statistically indistinguishable from the product form of Equation 11 for both links (maximum absolute deviation at Monte Carlo noise; χ^2 goodness-of-fit p -values of 0.06 and 0.23).

Hierarchical. The MCMC sampler of Section 2.11 recovers population means, covariances, and cut points on synthetic panels for both links, and its posterior uncertainty is calibrated: across 20 replications (Model B; 8 for Model A), 90% central posterior intervals cover the truth at 89% and 83% respectively, with 50% intervals at 51% and 43%. As an independent check of the hierarchy machinery, the forced-choice-only special case of our sampler agrees with an established hierarchical-MNL implementation on identical data to within 0.008 on every population parameter.

3.3 The Cost of Dichotomization

Collapsing the W -point response at a cut point yields a correctly specified binary dual-response model — dichotomization costs efficiency, never consistency (bias tests confirm this in every cell) — so the comparison is precision, computed two ways: exactly, from the Fisher information; and by paired Monte Carlo, fitting the ordinal model and the binary model to the *same* simulated datasets so that all differences are attributable to the coarsened second response.

Aggregate: modest at a lucky cut, material at an unlucky one. Table 2 reports exact variance ratios (equivalently, the extra sample the binary analyst needs) for the top-2-box convention and for the worst fixed cut, by configuration. Top-2-box dichotomization costs 4%–15% extra sample for the preference parameters while discarding 27%–52% of the second response’s own information; the worst cuts cost up to 33%. The analyst must commit to the cut before seeing the data; the ordinal model is uniformly within about 2% of the full-information benchmark. Aggregate demand-*prediction* ratios are small (1.01–1.07) because prediction variance is dominated by the forced-choice stage both estimators share — an honest caveat that the individual-level results below overturn.

Table 2: Exact asymptotic variance ratios versus the full 5-point model (extra sample required for equal precision), and the share of the second response’s information each coarsening retains. Top-2-box=the literature’s default dichotomization.

configuration	top-2-box, beta	worst cut, beta	omega share, top-2-box	omega share, worst
set1_moderate	1.041	1.091	0.73	0.38
set2_negative	1.088	1.119	0.59	0.40
set3_stress	1.152	1.327	0.59	0.19
set4_skewed	1.080	1.125	0.48	0.20

Source: [Article Notebook](#)

Individual level: the gaps widen. In the hierarchical setting — the model practitioners actually estimate — each respondent contributes only $T = 12$ ordinal reports, and the pooled cut points no longer rescue the coarsened second stage. Across 20 paired replications, top-2-box dichotomization inflates the variance of the shared cut point by a factor of 1.09, individual-level purchase-probability prediction error by 1.19 in variance terms, and individual part-worth RMSE by 4.2% — with identical forced-choice data. Holdout hit rates are statistically indistinguishable (0.511 vs. 0.510), as they should be: the first response carries the ranking information, and what dichotomization damages is precisely the demand margin the dual response exists to measure.

3.4 Specification Tests: Size and Power

Table 3 summarizes the operating characteristics of the tests from Section 2.9 over 300 (size) and 150 (power) replications per cell. Both tests hold their nominal 5% size, the estimators of λ and δ are unbiased, and two results deserve emphasis: the conditioning test rejects the DR-2Max second stage in *every* replication at both sample sizes — the 2Max-versus-inclusive-value question does not require large data to settle — and when the data truly come from a DR-2Max process, $\hat{\delta}$ lands on 1 (mean 1.00) and our model is rejected with power 100%, so the test is informative in both directions.

Table 3: Rejection rates of nominal-5% likelihood-ratio specification tests. Under the null (top rows), the lambda and delta tests hold size while ‘delta=1’ shows power against DR-2Max; under the alternatives (bottom rows), the targeted test rejects while the others stay quiet.

cell	n	reject lambda=1	reject delta=0	reject delta=1 (2Max)
size B	1200	0.067	0.050	1.000
size B	3600	0.063	0.027	1.000

Table 3: Rejection rates of nominal-5% likelihood-ratio specification tests. Under the null (top rows), the lambda and delta tests hold size while ‘delta=1’ shows power against DR-2Max; under the alternatives (bottom rows), the targeted test rejects while the others stay quiet.

cell	n	reject lambda=1	reject delta=0	reject delta=1 (2Max)
size A	3600	0.047	0.033	1.000
power lambda=0.7	3600	0.993	0.053	1.000
power lambda=1.3	3600	0.993	0.060	1.000
power lambda=0.7	1200	0.640	0.053	1.000
power 2max (delta=1)	3600	0.067	1.000	0.073
power 2max (delta=1)	1200	0.060	1.000	0.053

Source: [Article Notebook](#)

3.5 Misspecification Robustness

Four deliberate misspecifications, each fit with the (now wrong) standard model on hierarchical panels:

- **Wrong link.** Fitting Model B to Model A data and vice versa moves holdout purchase-probability predictions by at most 0.025 on the probability scale (versus 0.019 under the correct link), and WAIC selects the true link in 100% of replications at this design’s attractiveness spread. Link misspecification is a second-order concern for demand; it matters for extrapolation into the tails.
- **Ignored scale-usage heterogeneity.** With respondent-specific cut points in the data-generating process (dispersion 0.35 on c_{i1}), the common-cut model loses almost nothing: individual part-worth RMSE 0.532 versus 0.535 for the heterogeneous-cut model, with prediction and hit-rate differences equally negligible. At commercial task counts the extension of Section 2.7.1 is robustness insurance and a diagnostic, not a default — exactly what the identification discussion anticipated.
- **Non-Gumbel shocks.** With variance-matched normal first-stage errors, preference recovery and demand prediction are essentially unimpaired (individual RMSE 0.514, prediction RMSE 0.110); the misspecification is absorbed almost entirely by the cut-point locations (the top-2-box cut shifts by 0.63), which is where a structural-interpretation caveat belongs.
- **Task-order drift.** When the outside good drifts across the sequence (linear drift, and Hung-style probabilistic updating as in Hung, Kurz, et al. (2025)), the static model’s cut points absorb the average drift while part-worths remain accurate, and the simple order diagnostic — residual exceedance regressed on task position — detects the drift reliably (mean $|t| = 5.1$ under linear drift).

3.6 Designing the Scale

The information results of Section 2.10 become concrete design advice (Figure 1). With cut points at equal expected shares, a binary follow-up captures 71% (logistic link) or 46% (Gumbel link) of the continuous-report information limit; five points capture 95% and 78%; the increment beyond seven points is small. Cut *placement* matters as much as cut *count* for the Gumbel link: information-optimal cuts recover a further 15% over equal-share placement, whereas the logistic link is nearly indifferent — and label sets that produce the skewed usage typical of fielded purchase-intent scales cost 29% (Gumbel) and 4% (logistic) relative to optimal.

Discriminating between the two links, by contrast, is governed by the *attractiveness variation* of the design, exactly as Section 2.8 predicts. At this design’s natural inclusive-value spread, marginal-density comparisons select the true link in every replication from $n = 1,800$ tasks upward. On deliberately “squashed” designs — identical attribute variation, but the inclusive value equalized across tasks — the data carry *no* information about the link (either link fits the category frequencies exactly by relabeling cut points), and selection degenerates into an artifact of the curvature correction: the same link is chosen essentially always, regardless of the truth ($P(\text{correct} \mid A \text{ true}) = 0.00$, $P(\text{correct} \mid B \text{ true}) = 0.97$ at $n = 7,200$). Studies intending to test the behavioral story must engineer high- and low-attractiveness tasks; studies that only need demand can ignore the distinction.

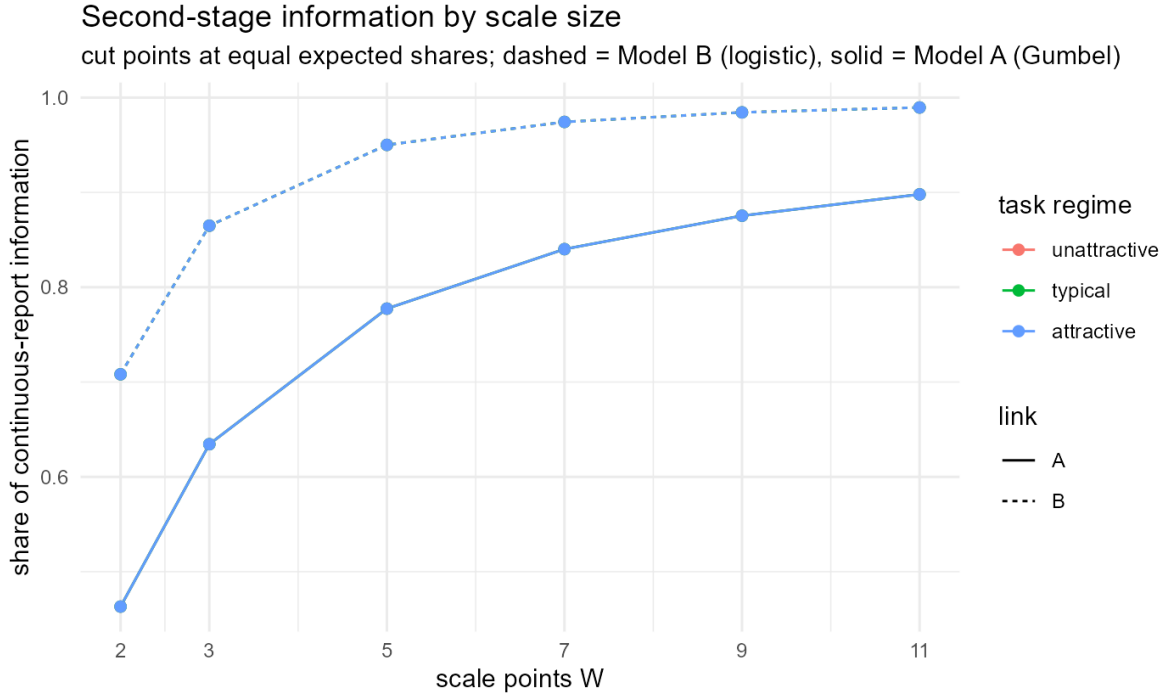


Figure 1: Share of the continuous-report information captured by a W -point second response (cut points at equal expected shares). The binary response is the least informative member of the family; most of the gap closes by five points.

Source: [Article Notebook](#)

3.7 Economic Stakes: A Pricing Counterfactual

Finally, we push each estimator through the decision it exists to inform. A synthetic market — five brands, a quality attribute, wide price variation, heterogeneous preferences calibrated to own-price elasticities of about -1.9 — generates $W = 5$ dual-response data ($N = 500$, $T = 12$); each estimator's demand system then sets Nash equilibrium prices (damped best responses; multi-start verified). Demand is defined identically for all systems as top-2-box purchase, so the well-placed binary estimator is answering exactly the right question; posterior-predictive demand uses full parameter draws.

Table 4: Nash-pricing distortions by estimator: mean and max absolute equilibrium-price error versus the true equilibrium; error in the system’s own profit forecast at its equilibrium; and realized profit forgone relative to pricing at the true equilibrium.

system	mean price err %	max price err %	profit forecast err %	profit forgone %
ordinal	1.11	2.13	-2.98	0.37
bin_t2b	1.10	2.38	-2.47	-0.31
bin_top	2.57	5.73	-59.29	-0.84
heur_juster	4.84	8.52	105.82	2.18
heur_t2b	4.84	8.52	52.94	2.18

Source: [Article Notebook](#)

Three lessons (Table 4). First, the full ordinal model prices the market almost exactly. Second, the *well-placed* binary dual response also prices well — consistent with the efficiency analysis — but the analyst does not get to choose the placement after the fact: dichotomizing at the top box misprices by up to 5.7% and misstates its own profit forecast by 59%, because a top-box model can only answer a top-box question. Third, the constant-probability heuristic is the expensive shortcut: its purchase incidence cannot respond to price, so it underprices systematically (its equilibrium is the inside-good-only equilibrium — the calibration constant cancels from every best response) and overstates forecast profits by 106%. The heuristic’s failure is structural, not statistical: no amount of data fixes it.

4 Empirical Application

We illustrate the model with data from a commercial choice-based conjoint study conducted in partnership with a consumer insights consultancy, in which each choice task was followed by an ordinal purchase-likelihood question.

4.1 Data

4.2 Model Specifications and Estimation

We estimate six specifications on the same data: Model A and Model B (Section 2), each in aggregate and hierarchical form; the binary dual-response benchmark applied to the dichotomized scale (top-two-box); and the constant-probability heuristic. Hierarchical specifications use the sampler of Section 2.11.

Pre-specified test battery. Before any model comparison, we commit to the three specification tests of Section 2.9 on the aggregate fit: the unit-slope test ($\lambda = 1$), the conditioning test

($\delta = 0$), and the DR-2Max test ($\delta = 1$) — whose operating characteristics are established in Section 3.4 — together with the A-versus-B link comparison by holdout likelihood and WAIC. We also pre-commit to re-running the pricing counterfactual of Section 3.7 on the empirical posteriors, replacing the synthetic market with the study’s own category structure.

4.3 Results

4.4 Scale-Usage Heterogeneity

5 Discussion

5.1 What the Behavioral Model Changes

The premise that the consumer observes everything about the inside goods but not the outside-good shock does more than motivate a likelihood; it reorganizes how the dual-response literature’s models relate to one another. In the binary literature, the choice between DR-2Max and DR-AnyMax was framed as a question about the second question’s wording — whether the respondent evaluates her selected option or the whole set (Diener et al. 2006) — and the shared-error Unified Model of Hung, Liu, et al. (2025) was motivated by the internal consistency of a single utility draw. Our framework arrives at the inclusive-value form from the primitive assumption about information: because the maximum of Gumbel utilities is independent of which good attains it, *any* coherent report about the best option’s standing against the outside good must depend on the task only through its inclusive value. The empirical success of the AnyMax/Unified form over DR-2Max, documented in both Diener et al. (2006) and Hung, Liu, et al. (2025), is what this framework predicts. Where our account differs from Brazell et al. (2006) is in what the second response *is*: not a second choice from an augmented set, but a report about an unresolved comparison — which is what licenses asking for it on an ordinal scale in the first place.

The distinction between Model A and Model B is, likewise, a substantive behavioral question dressed as a link function. Model A takes the respondent’s uncertainty literally — she reports a probability — while Model B assumes she resolves the outside-good draw and grades the outcome. Their binary special cases already separate the two: Model B recovers the standard logit second stage, while Model A implies a complementary log-log second stage even for yes/no data. Because the two links diverge mainly in the tails, discriminating between them requires either large samples or designs with wide variation in task attractiveness; the model comparison in Section 4 provides evidence from one commercial dataset, but we regard the question as open and worth testing on the many existing binary dual-response datasets, where the Model A binary case can be estimated with a one-line change to current code.

5.2 Implications for Practice

Four points bear directly on applied work. First, if an ordinal dual response has been fielded, dichotomizing it is costly and arbitrary: the analyst’s choice of cut is made before seeing the data, the penalty ranges from a few percent of sample at a lucky cut to a third of it at an unlucky one (Section 3.3), the damage concentrates at the individual level where hierarchical estimation lives, and a dichotomized model can answer questions only at its own threshold — the pricing counterfactual’s top-box analyst misstates the profit forecast by more than half for exactly that reason. Second, constant-probability weighting has a *structural* flaw, not a statistical one: beyond the set-dependence of scale-point meaning (Equation 7 and Table 1), a fixed purchase incidence cancels out of every pricing first-order condition, so the heuristic’s price recommendations ignore the no-buy margin entirely — no amount of data fixes this. Third, the information results in Section 2.10 and Section 3.6 give concrete design guidance: five scale points capture most of the binary-to-continuous information gap (about 95% under the logistic link, 78% under the Gumbel link, versus 71% and 46% for binary), gains beyond seven points are small, and — for the Gumbel link especially — *where* the labels place the cut points matters as much as how many there are. Fourth, studies that intend to test the behavioral model (the link, the unit slope) should engineer variation in task attractiveness; designs without it cannot discriminate the links even in principle (Section 2.8), a fact our squashed-design experiment confirms empirically.

5.3 Relation to the Outside-Good Literature

Our model holds the outside good’s *mean* utility fixed at zero and locates all action in the consumer’s uncertainty about its realization. Two adjacent strands suggest natural extensions. Hung, Liu, et al. (2025) show that a budget constraint applied to the second response — but not the forced choice — improves fit and materially changes equilibrium price predictions; the same asymmetric treatment is available here, censoring the ordinal report for options priced above a respondent-level budget, and the ordinal response should sharpen the identification of that budget since near-threshold options will earn lukewarm rather than merely negative reports. Hung, Kurz, et al. (2025) document that the no-choice option’s utility can drift over the course of a survey as respondents learn about the category; in our notation this is drift in the cut points (or in the outside-good location), and the task-order diagnostics of Section 4.1 carry over directly. More broadly, the opt-out literature’s warning that a single constant rarely captures heterogeneous non-purchase behavior (Campbell and Erdem 2019; Haaijer et al. 2001) applies to the dual response as well; the heterogeneous-cut-point extension of Section 2.7.1 is one response, and richer structures (e.g., serial non-purchasers as a latent class) are compatible with the framework.

5.4 Limitations

Four limitations qualify the results. First, the Gumbel error assumption is load-bearing: the factorization of the joint likelihood and the inclusive-value form of the second response both rest on max-stability, and relaxing the error family (e.g., toward mixed or nested structures within a task) would break the closed form, though not the framework — the second response would simply require simulation over the max distribution. Our robustness experiments soften the concern in one direction: with variance-matched normal shocks, preference and demand estimates are essentially unimpaired, the misspecification being absorbed by the cut-point locations — so structural interpretation of the cut points deserves more caution than the demand quantities built from them. Second, we treat tasks as independent draws, abstracting from learning, fatigue, and reference formation across the sequence (Hung, Kurz, et al. 2025); the task-order diagnostic of Section 3.5 detects such drift reliably when present. Third, the common-cut-point model assumes a shared reading of the scale labels; the extension of Section 2.7.1 relaxes this, but individual-level identification is thin at commercial task counts, and our simulations find that ignoring moderate scale-usage heterogeneity costs little — the extension earns its keep as insurance and as a diagnostic, not as a default. Fourth, like all stated-preference methods, the model calibrates stated rather than revealed purchase behavior; the ordinal report inherits whatever hypothetical bias attends purchase-intent questions generally (Morrison 1979), and linking the estimated cut points to realized purchase rates — the ordinal analogue of the calibration literature on binary intentions — is a natural next step. A separate methodological caution, documented in Appendix C, is that harmonic-mean marginal-likelihood estimates are optimistic; we therefore anchor model comparisons in holdout likelihood and WAIC alongside the conventional estimator.

6 Conclusion

Dual-response conjoint designs separate a question consumers find easy — which of these do you like best? — from one they find genuinely uncertain — would you actually buy it? This paper takes that uncertainty seriously as a modeling primitive. When the consumer observes everything about the displayed alternatives but not her valuation of the outside good, the natural follow-up question is ordinal, and the natural model is the one derived here: a multinomial logit for the forced choice joined to a cumulative-link ordinal model on the task’s inclusive value, with the link function determined by whether the respondent reports her purchase probability directly (Model A) or grades a resolved comparison (Model B). The framework nests the leading binary dual-response models as its two-point special case, so nothing is lost relative to current best practice; a W -point scale strictly adds Fisher information, and the common practitioner shortcuts — dichotomizing the scale or assigning fixed purchase probabilities to its points — are shown to be, respectively, wasteful and internally inconsistent. Simulation evidence confirms that the aggregate and hierarchical Bayesian estimators recover

the model's parameters, including the cut points that give the scale labels quantitative meaning, and the empirical application demonstrates the model on commercial data.

Beyond the specific model, the paper's broader point is that response formats and statistical models should be developed together. The ordinal purchase-likelihood question earned its place in practice because it matches how respondents experience purchase decisions; it lacked only a likelihood. Supplying one turns an informal fielding habit into a measurement instrument — and opens a research agenda that includes calibrating estimated cut points against realized purchase behavior, exploiting the ordinal response to identify budget constraints, and optimal design of the scale itself.

Appendix A: Proofs

A.1 Proof of Lemma 2.1

Let $u_j = V_j + \eta_j$ with $\eta_j \stackrel{iid}{\sim} \text{Gumbel}(0, 1)$ for $j \in \mathcal{J}^+$, so u_j has CDF $F_j(t) = \exp(-e^{-(t-V_j)})$ and density $f_j(t) = e^{-(t-V_j)} \exp(-e^{-(t-V_j)})$. For any j and any u ,

$$\Pr(j^* = j, u^* \leq u) = \int_{-\infty}^u f_j(t) \prod_{k \neq j} F_k(t) dt = \int_{-\infty}^u e^{V_j} e^{-t} \exp(-e^{-t} S) dt,$$

using $\prod_{k \neq j} F_k(t) = \exp(-e^{-t} (S - e^{V_j}))$ with $S = \sum_k e^{V_k}$, and collecting terms. Substituting $s = e^{-t} S$ (so $ds = -e^{-t} S dt$) gives

$$\Pr(j^* = j, u^* \leq u) = \frac{e^{V_j}}{S} \int_{e^{-u} S}^{\infty} e^{-s} ds = \underbrace{\frac{e^{V_j}}{S}}_{\Pr(j^*=j)} \times \underbrace{\exp(-Se^{-u})}_{F_{\text{Gumbel}(\bar{\mu}, 1)}(u)}.$$

Letting $u \rightarrow \infty$ delivers the MNL choice probability; summing over j delivers $u^* \sim \text{Gumbel}(\bar{\mu}, 1)$ with $\bar{\mu} = \ln S$; and the factorization of the joint law for every (j, u) is precisely the independence of j^* and u^* . $\square$

A.2 Derivation of the information increment (Equation 18)

The second response contributes the multinomial log-likelihood $\ell = \sum_w \mathbb{1}\{y = w\} \ln \pi_w$ with $\pi_w = G(c_w - \bar{\mu}) - G(c_{w-1} - \bar{\mu})$. Since $\partial \bar{\mu} / \partial \boldsymbol{\beta} = \sum_j \Pr(j^* = j) \mathbf{x}_j = \bar{\mathbf{x}}$, the score with respect to $\boldsymbol{\beta}$ in category w is $-[g_w - g_{w-1}] / \pi_w \cdot \bar{\mathbf{x}}$, writing $g_w = g(c_w - \bar{\mu})$ and $g = G'$ (with $g_0 = g_W = 0$). The information is the expected outer product of the score:

$$\mathcal{I}^{(2)} = \sum_{w=1}^W \pi_w \left(\frac{g_w - g_{w-1}}{\pi_w} \right)^2 \bar{\mathbf{x}} \bar{\mathbf{x}}' = \left[\sum_{w=1}^W \frac{(g_w - g_{w-1})^2}{\pi_w} \right] \bar{\mathbf{x}} \bar{\mathbf{x}}' = \omega(\bar{\mu}; \mathbf{c}) \bar{\mathbf{x}} \bar{\mathbf{x}}'. \quad \square$$

The joint information over $(\beta, \mathbf{c})$, including the cut-point blocks used by the estimation code, follows from the same score with $\partial\pi_w/\partial c_m = g_w \mathbb{1}\{m = w\} - g_{w-1} \mathbb{1}\{m = w-1\}$.

A.3 Monotonicity of ω under refinement

Let $\mathcal{C}'$ be a coarsening of the partition $\mathcal{C}$ obtained by merging adjacent categories w and $w+1$. The merged cell contributes $(g_{w+1} - g_{w-1})^2 / (\pi_w + \pi_{w+1})$ to ω' , while the two original cells contribute $(g_w - g_{w-1})^2 / \pi_w + (g_{w+1} - g_w)^2 / \pi_{w+1}$ to ω . Writing $a = g_w - g_{w-1}$, $b = g_{w+1} - g_w$, $u = \pi_w$, $v = \pi_{w+1}$, the Cauchy–Schwarz inequality in Engel form gives

$$\frac{(a+b)^2}{u+v} \leq \frac{a^2}{u} + \frac{b^2}{v},$$

so each merge weakly decreases ω , and any coarsening — being a finite sequence of adjacent merges — does the same. The binary response is the coarsest member of the family and hence the least informative; conversely, as the mesh of the partition goes to zero the interval score converges to the exact-observation score and $\omega \rightarrow \int (g')^2 / g \, dz$, the Fisher information of the latent’s location (equal to 1 for the standard Gumbel and 1/3 for the standard logistic). $\square$

Appendix B: Estimation Details

Aggregate model. The likelihood is maximized over the working parameters $(\beta, c_1, \ln(c_2 - c_1), \dots, \ln(c_{W-1} - c_{W-2}))$, which enforce cut-point monotonicity without constraints; standard errors for the cut points follow by the delta method. Interval probabilities are computed with tail-stable expressions $(\Lambda(h)\Lambda(-l)(1 - e^{l-h}))$ for the logistic link and $e^{-e^{-h}}(1 - e^{-(e^{-l} - e^{-h})})$ for the Gumbel link) so that extreme cut points do not lose precision to cancellation.

Hierarchical model. The sampler is a random-walk Metropolis-within-Gibbs in the style of Rossi et al. (2005) with three blocks per iteration: (1) all respondents’ β_i (and, in the heterogeneous-cut-point extension, θ_i) are updated simultaneously by independent normal random-walk proposals with covariance $s_i^2 \Sigma$; (2) the common cut-point block is updated by a random-walk proposal on the unconstrained scale; (3) the population moments $(\bar{\beta}, \Sigma)$ are drawn from their conjugate normal and inverse-Wishart conditionals. Priors: $\bar{\beta} \sim \mathcal{N}(\mathbf{0}, 100I)$, $\Sigma \sim \text{IW}(\text{dim}+3, (\text{dim}+3)I)$, and $\mathcal{N}(0, 100I)$ on the transformed cut points. Step sizes are initialized at $2.93/\sqrt{\text{dim}}$ and adapted only during burn-in toward acceptance rates of 0.23 (individual blocks) and 0.30 (cut-point block). Unless noted, we run 30{,}000 iterations, discard 10{,}000, and keep every 10th draw.

The sampler was validated four ways (replication code accompanies the paper): parameter recovery on synthetic panels for both links; repeated-sampling coverage of 50% and 90% posterior intervals at nominal rates; agreement of the forced-choice-only special case with an

independent hierarchical-MNL implementation (posterior means matching to less than 0.01 on all hyperparameters on common data); and convergence of the posterior toward the aggregate MLE when the data are generated without heterogeneity.

Appendix C: Marginal-Likelihood Estimation

Model comparisons in the paper report the log marginal density (LMD). For hierarchical models we follow the standard practice in this literature (Hung, Liu, et al. 2025) and use the harmonic-mean estimator of Newton and Raftery (1994), complemented by two quantities that do not rely on it: the WAIC computed from pointwise per-respondent likelihoods (Watanabe 2010; Vehtari et al. 2017) and the log posterior-predictive density of held-out tasks. The reason for the complements is that the harmonic-mean estimator is optimistic. In the aggregate model — where the parameter space is small enough for exact methods — we can quantify this directly: bridge sampling (Meng and Wong 1996; Gronau et al. 2020) and a Laplace approximation agree with each other to well under one log point, while the Newton–Raftery estimate exceeds them by dozens of log points, and trimming the smallest-likelihood draws widens rather than closes the gap:

Table 5: Aggregate-model log marginal density by estimator (simulated data, $n = 3,600$ tasks; from R/hb_validation.R).

estimator	log marginal density	vs. bridge
bridge	-9844.14	0.00
laplace	-9844.15	-0.01
NR	-9801.78	42.36
NR trim 5%	-9799.63	44.51
NR trim 20%	-9798.42	45.73

Source: [Article Notebook](#)

Because the bias is similar for models of similar dimension, Newton–Raftery *rankings* between the ordinal and binary dual-response models are typically preserved — but we treat the WAIC and holdout comparisons as primary whenever the two disagree.

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